

EDUCATION

Bachelor of Technology (B.Tech), Computer Science & Engineering

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar (CGPA: 8.1 upto 5th semester)

Expected Graduation: 2027

Relevant Coursework: Software Engineering, Database Management Systems, Computer Organisation and Architecture, Machine Learning, Artificial Intelligence, Cloud Computing, Computer Intelligence, Computer Networks, Data Structures and Algorithms

SKILLS

Languages and Databases: Python, Java, C, MySQL

Frameworks/Libraries: PyTorch, Tensorflow, NumPy, Pandas, Scikit-learn, Matplotlib, OpenCV, Gymnasium

Web: HTML, CSS, JavaScript

Tools: Git, VS Code, Kaggle, Jupyter

WORK EXPERIENCE

Nebula Space Organisation | Bhubaneswar, India

Dec 2023 - Feb 2025

Research & Development Intern:

- Automated data extraction pipelines in Python processing structured data across multiple spreadsheets, reducing manual effort for internal reporting
- Conducted systematic review of 50+ technical papers on satellite subsystems, synthesizing material selection criteria for specific engineering use cases

PROJECTS

Adaptive Image Compression using CNN-based Importance Maps

Python, OpenCV, NumPy, PyTorch, DCT

- Developed adaptive image compression using CNN importance maps with a JPEG (DCT) pipeline and block-wise adaptive quantization
- Outperformed standard JPEG by +0.14 dB PSNR (34.55 vs 34.41 dB) at comparable bitrate, validated across 137 held-out test images at Q=40
- Achieved 9x compression ratio (192 KB → 21 KB) while maintaining higher perceptual quality than standard JPEG

Deep Reinforcement Learning – Atari Ms. Pacman (DQN)

Python, PyTorch, Gymnasium (Atari), NumPy, OpenCV

- Built a DQN agent in PyTorch trained directly on raw pixel input (64×64) with 4-frame stacking for temporal representation, no handcrafted features
- Achieved ~54% improvement in average reward across training, with evaluation scores of 500–700 and best episode exceeding 1,400+ points
- Conducted ablation study on frame stacking, quantifying its impact on the agent's ability to infer motion and directional context

Comparative Analysis of Deep Q-Network Variants for LunarLander-v3

Python, PyTorch, Gymnasium (Box2D), NumPy

- Compared four DQN variants in PyTorch across 1,500 episodes on LunarLander-v3
- Vanilla DQN achieved Avg100: 233, converging ~400 episodes faster than all variants
- Dueling DQN showed critical instability at ep. 850–900, highlighting that architectural complexity doesn't guarantee improvement on well-shaped reward environments

Advanced MobileNet-SSD Based Object Recognition System for Visually Impaired Assistance

Python, TensorFlow, OpenCV, MobileNet-SSD, Computer Vision, Text-to-Speech

- Built a real-time object detection system using MobileNet-SSD with ~71% accuracy in live environments
- Optimized performance using SSIM-based frame skipping, reducing redundant computation
- Integrated text-to-speech feedback with duplicate filtering for improved user experience

Continuous Control for Simulated Driving using PPO and SAC (Feb 2026 - Present)

Python, PyTorch, Gymnasium, NumPy

- Developing and training Proximal Policy Optimization (PPO) and Soft Actor-Critic (SAC) agents for continuous-control tasks in a simulated environment
- Exploring training stability, convergence behavior, and reward optimization across policy-gradient methods

COURSES AND CERTIFICATIONS

- Artificial Intelligence: Agentic AI, Generative AI, Reinforcement Learning – Udemy
- CS50's Introduction to Artificial Intelligence with Python (HarvardX) - edX
- Data Science: Machine Learning (HarvardX) - edX